ZEERBOX · TRACEABILITY MATRIX

# ZeerBox traceability matrix

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## DOCUMENT CONTROL

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| Document ID | ZBX-RTM-001 | Version | 0.3 |
| Status | Draft | Date | 2026-10-25 |
| Author | Amish Chadha | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept | | |
| Repository | github.com/BoujeeEnjinia1701/zeerbox | | |
| Source | docs/03b-traceability.md @ 05b0b75 | | |

## REVISION HISTORY

| Version | Date | Author | Change |
|---|---|---|---|
| 0.3 | 2026-10-25 | Amish Chadha | Generated by .kit/trace.py |
| 0.2 | 2026-10-25 | Amish Chadha | Generated by .kit/trace.py |
| 0.1 | 2026-10-25 | Amish Chadha | Generated by .kit/trace.py |

# ZeerBox traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 14 user requirements, 12 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

| User requirement | Product requirement | Title | Evidence | Verification | Status |
|---|---|---|---|---|---|
| UR-001, UR-002 | PR-001 | Storage capacity and reach | ZBX-CAL-001, ZBX-BLD-001 | Analysis of the model (CAL-001 s2), TRL 3; loading trial, TRL 4 | Met on paper: 24 crates, 480 kg; top shelf 1.40 m |
| UR-003 | PR-002 | Cooling at the design point | ZBX-CAL-001, ZBX-BLD-001, ZBX-STM-001, ZBX-OPT-001 | Psychrometric and heat balance (CAL-001 s4), TRL 3; logged field data, TRL 4 | Met on paper: 10.7 K (9.8 to 11.4 K); store 24.3 °C |
| UR-004 | PR-003 | Humidity for produce | ZBX-CAL-001, ZBX-BLD-001, ZBX-OPT-001 | Calculation (CAL-001 s5), TRL 3; logged field data, TRL 4 | At risk: 81 % (77 to 85 %) |
| UR-005 | PR-004 | Humid-weather detection and mode lamp | ZBX-CAL-001, ZBX-BLD-001, ZBX-OPT-001 | Logic review (CAL-001 s6), TRL 3; bench test with sensors, TRL 4 | Met by logic review: 30 °C and 75 % RH gives 3.7 K and switches to hold; sensing error 0.5 K (STM S32) |
| UR-006 | PR-005 | Mold and air-quality guard | ZBX-CAL-001, ZBX-BLD-001 | Logic review (CAL-001 s6), TRL 3; bench test, TRL 4 | Met by logic review: hold mode about 8 air changes an hour at about 2 W; with no power for more than about 40 h the fans stop (STM S36, ZBX-DEC-001 D3) |
| UR-007 | PR-006 | Shelf life | ZBX-CAL-001 | Literature comparison, TRL 3; paired storage trial with the partner, TRL 4 | To verify at TRL 4: ZECC data (about 1.8 times) come from a cooler, more humid chamber |
| UR-008 | PR-007 | Solar operation | ZBX-CAL-001, ZBX-BLD-001 | Energy balance (CAL-001 s7), TRL 3 | Met on paper: 525 Wh from the 150 W panel (41 % margin) |
| UR-009 | PR-008 | Water use and sump | ZBX-CAL-001, ZBX-BLD-001, ZBX-STM-001 | Water balance (CAL-001 s8), TRL 3 | Met on paper: about 43 L (no cavity water; 11 L bleed, ZBX-DEC-001 D5); 60 L sump for 43 L |
| UR-010 | PR-009 | Low voltage and safe | ZBX-CAL-001, ZBX-BLD-001, ZBX-STM-001 | Design review (CAL-001 s9), TRL 3 | Met by design review: 15 A battery fuse on 2.5 mm² leads, 5 A load fuse on the 1.5 mm² load circuit, both DC rated (ZBX-DEC-001 D1, D5) |
| UR-011, UR-012 | PR-010 | Local build and repair | ZBX-CAL-001 | Design review, TRL 3; parts survey and swap trials with the partner, TRL 4 | To verify at TRL 4 |
| UR-013 | PR-011 | Durability | ZBX-CAL-001, ZBX-BLD-001 | Material review and structural checks (CAL-001 s10), TRL 3; | To verify at TRL 4; roof footings 1.8 times the uplift at |

| User requirement | Product requirement | Title | Evidence | Verification | Status |
|---|---|---|---|---|---|
| | | | | supplier data and field life, TRL 4 | 30 m/s; beams and purlins sized |
| UR-014 | PR-012 | Affordable kit | ZBX-CAL-001, ZBX-BLD-001 | Priced BOM ( `bom/bom.csv` ), CAL-001 s11, TRL 3 | At the target: kit USD 300; structure USD 574, reported separately |

## Gaps

None.